Predictive Analytics for Smart City Infrastructure: Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG

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***Abstract*—Urban traffic management needs forecasts that are accurate at several horizons, honest about their uncertainty, and explainable enough to justify an intervention. This paper presents an end-to-end research prototype for Manhattan that couples a context-gated spatiotemporal graph transformer for 15/30/60-minute link-speed forecasting with a constrained, retrieval-augmented agentic workflow that produces simulation-only signal-timing advisories. The forecaster consumes a directed road-link graph built from NYC DOT real-time speed links as a PyTorch Geometric Temporal signal, fuses weather, 311 and calendar context through an availability-aware gate, and is calibrated with split conformal prediction. Explanations are computed on the trained model with grouped KernelSHAP. Advisories are drafted by a locally hosted large language model that must cite passages retrieved from an allowlisted corpus of federal signal-timing guidance and a project-defined policy, are checked by deterministic safety constraints, and are evaluated against a baseline controller in SUMO with matched seeds before being queued for human review. All data are public and all reported figures are generated from stored artifacts of one end-to-end run on nine months of 2026 NYC DOT speeds (16 usable links). On the 45-day test partition the proposed model reaches an MAE of 4.77, 5.33 and 6.15 mph at 15, 30 and 60 minutes, 7–16 % below persistence and below both tree baselines, but within 0.05 mph of a GRU and a temporal-only Transformer; the availability gate improves MAE by 0.07 mph (bootstrap interval excluding zero) while the operational weather features worsen it by 0.05 mph, a negative result. Conformal intervals reach 0.90–0.91 coverage overall but 0.76–0.80 under congestion, which the deterministic gate converts into a 26 % deferral rate. Retrieval removes fabricated citations (3 of 3 without it, 0 of 9 with it), and the constraint layer, not the model, decides admissibility; no advisory reached the simulation gate because the model chose monitoring on the mapped corridor. A first 45-run SUMO controller comparison is superseded: an audit showed that SUMO had re-routed half of the corridor traffic off West Street, and after correction neither the generated nor Webster re-timed signal programs let the modelled corridor serve the observed West Street volume within the preregistered baseline test, so the corrected comparison is not run. Negative, partial and not-run outcomes are reported in Section VII exactly as produced by the pipeline.**

***Index Terms*—traffic forecasting, graph neural networks, transformers, conformal prediction, explainable AI, retrieval-augmented generation, agentic systems, traffic simulation, smart cities**

I. Introduction

Urban traffic-management centres typically observe congestion as it happens. Linking an *advance* forecast to an explainable, evidence-grounded and tested operational recommendation remains difficult: traffic depends on recurrent demand, neighbouring links, weather, planned events, disruptions and infrastructure condition, and a forecast alone does not establish which intervention helps. An LLM-generated signal recommendation is, by itself, not evidence that an action is safe or effective.

We study a decision-support pipeline for the scenario of a permitted event near a busy corridor in which rain and a nearby collision coincide. The pipeline (i) forecasts link speeds 15, 30 and 60 minutes ahead with a context-gated graph spatiotemporal Transformer, (ii) attaches split-conformal prediction intervals, (iii) explains the forecast of the actual model with grouped Shapley values, (iv) retrieves operational guidance with a FAISS index, (v) asks a local LLM (Qwen2.5-7B-Instruct via Ollama) for schema-constrained candidate actions, (vi) applies deterministic safety constraints, and (vii) tests admissible candidates in a SUMO microscopic simulation before human review.

Proposed contributions (stated as proposals; their effects are reported only where measured): (1) availability-aware context fusion in the graph forecaster; (2) an explicit uncertainty-and-evidence gate that decides whether advisory generation proceeds; (3) a simulation-based candidate gate with side-road protection; and (4) a traceable evaluation that links forecast quality, grounding, deferral and simulated intervention outcomes to versioned artifacts. We do not claim deployment, real-world congestion reduction or causal explanations.

II. Related Work

Graph traffic forecasting. Diffusion-convolutional recurrent networks [1], spatio-temporal graph convolution [2] and Graph WaveNet [3] established graph neural networks for sensor-network speed forecasting; attention-based variants use graph attention [4] and Transformer encoders [5]. Our model combines GATv2 message passing with a temporal Transformer

and adds an availability-aware gate over feature groups; we treat it as a project-designed architecture, not a new model family.

Uncertainty. Split conformal prediction [6], [7] gives finite-sample marginal coverage under exchangeability; for time series this assumption can fail, motivating adaptive and non-exchangeable variants [8], [9]. We therefore report empirical coverage by horizon and scenario.

Explainability. SHAP [10] attributes predictions to inputs; for tree baselines we use TreeSHAP, and for the graph model a grouped KernelSHAP over feature groups and graph neighbourhoods.

Retrieval-augmented LLMs. RAG [11] conditions generation on retrieved passages; we embed passages with Sentence-BERT models [12] and search them with FAISS [13]. The LLM is Qwen2.5 [14].

Simulation. SUMO [15] with TraCI is widely used to evaluate signal control. We use it as a scenario-based testbed and do not claim calibration to real demand.

III. Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed graph whose nodes are monitored directed road links and whose edges connect a link to links that can receive its traffic. At each 5-minute step t every node v carries a feature vector $x_{t,v}$ partitioned into groups $g \in \{\text{traffic, calendar, weather, infrastructure, events, incidents}\}$ with availability indicators $m_{t,v,g}$. Given the last $L = 24$ steps, the task is to predict observed link speed $y_{t+h,v}$ for $h \in \{3, 6, 12\}$ steps (15/30/60 min), together with an interval $[\hat{y} - q_h, \hat{y} + q_h]$. A downstream advisory is admissible only if data quality, interval width, retrieved evidence, deterministic constraints and a simulation comparison all pass; otherwise the system returns a distinct DEFER or REJECT status. Training minimises a masked Huber loss over observed targets only.

IV. Data

All sources are public (Table I). Speed targets come from the NYC DOT Real-Time Traffic Speed Data feed (Socrata `i4gi-tjb9`), restricted to Manhattan links with geometry, for 2026-01-01 to 2026-10-20. Each record carries a floating local timestamp that is localised to `America/New_York` and converted to UTC; the ambiguous daylight-saving hour is nulled rather than guessed. Records are snapped to a right-labelled 5-minute grid per link, averaged within a bin, bounded to physically plausible speeds, and forward-filled causally for at most three steps. Context comes from Open-Meteo (historical forecasts for the operational track, the finalised archive for the retrospective track), NYC 311 service requests with a declared 24-hour lag, Motor Vehicle Collisions (retrospective only; the feed lagged publication by about four months at download time) and NYC permitted events. Sampled automated traffic volume counts parameterise simulation demand only. OpenStreetMap supplies geometry for the simulated corridor. Every download is logged with URL, time, HTTP status and SHA-256, and a server-side audit of dataset identity and observed date ranges precedes every download; the full-year link enumeration timed out server-side, so the link set is the 125 links citywide active since 2026-09-20, of which 26 are in Manhattan. Of 1,968,303 raw Manhattan speed records, 978,018 carry a zero speed (the feed's no-measurement encoding) and are treated as missing, never imputed; 990,226 valid records fill a 84,084-step grid over 19 links (observed fraction 0.611), seven links being dead feeds with no valid 2026 record. Links with at least 50 % training-period coverage enter the models: 16 links, three of which have no valid test-period observation and are reported as NOT_EVALUATED (`reports/data_audit/link_coverage_operational.csv`, `reports/experiments/link_test_support_operational.`

V. Proposed Methodology

A. Context-gated GATv2–Temporal Transformer

For each node and time step, every feature group g is projected to a common width $d = 64$, $h_{t,v,g} = \mathrm{ReLU}(W_g x_{t,v,g})$. A scoring network gives $s_{t,v,g} = w^\top \tanh(U h_{t,v,g})$; scores of unavailable groups ($m_{t,v,g} = 0$) are masked to $-\infty$ and a softmax over the remaining groups yields weights $\alpha_{t,v,g}$. The fused representation is $z_{t,v} = \sum_g \alpha_{t,v,g} h_{t,v,g}$; if no group is available a learned "unavailable" vector is used and the event is logged. Two GATv2 layers [4] (4 heads, residual connections, layer normalisation) propagate z over the fixed directed graph at every time step (implemented with PyTorch Geometric [16]). Each node's sequence then receives a learned positional embedding and passes through a 2-layer, 4-head Transformer encoder (feed-forward width 128, dropout 0.1); a linear head on the last state outputs the three horizons. Targets are standardised per link with training-period statistics and the masked Huber loss is averaged over observed targets only; batches without any observed target are skipped and counted. Optimisation: AdamW (lr 10^{-3}, weight decay 10^{-4}), batch size 8, gradient clipping at 1.0, early stopping on validation MAE with patience 5 and at most 30 epochs, seed 42. The temporal snapshots are held in a PyTorch Geometric Temporal [17] `StaticGraphTemporalSignal`, from which training windows are drawn lazily. The gate weights are not interpreted as causal importance.

B. Baselines and ablations

Persistence (last observation), a training-period historical average per link, day type and 5-minute slot, XGBoost and LightGBM (one model per horizon on 12 speed lags and the same causal features), a GRU, and a temporal-only Transformer (the proposed model without message passing). Ablations remove the gate (concatenation instead), or one feature group at a time.

C. Uncertainty and reliability gate

With the selected model frozen, per-horizon absolute residuals on the calibration partition give the split-conformal quantile q_h as the $\lceil (n+1)(1-\alpha) \rceil$-th smallest residual ($\alpha = 0.1$); a per-link variant is also computed when at least 50 calibration

residuals exist. The advisory gate defers when data are older than 20 minutes, more than one third of recent traffic inputs are missing, the interval is wider than a threshold chosen on calibration data (80th percentile of per-link widths), or calibration support is below 200 samples.

D. Explanations

For an alert (link v, horizon h, origin t) the explained scalar is the predicted speed in mph. Players are the target's recent traffic (last 30 min), older traffic (30–120 min), the traffic of its graph neighbours and each context group; an absent player takes values from 16 training-period background windows. All 2^P coalitions are evaluated (exact KernelSHAP [10]). We additionally measure neighbour-edge removal and temporal-window sensitivities. Tree baselines are explained with TreeSHAP and labelled as baseline explanations.

E. Retrieval, LLM and agentic workflow

Approved documents (U.S. federal signal-timing guidance, SUMO documentation, and a clearly labelled project-defined experimental policy) are hashed, allow-listed, chunked with a 200 word-piece target and indexed with a FAISS inner-product index over normalised all-MiniLM-L6-v2 embeddings; chunks that still exceed the encoder's 256-token window (171 of 1,633 in the built index, mostly tables of the PDF manual) are truncated by the encoder, and the count is recorded in the index manifest. A query whose best passages score below 0.35 returns INSUFFICIENT_EVIDENCE. The LLM receives forecasts, intervals, quality flags, computed attributions, retrieved passages (marked as untrusted text) and the allowed action catalogue, and must return JSON that satisfies a Pydantic schema enforced through Ollama's structured-output option. A LangGraph state machine runs quality $\rightarrow$ forecast $\rightarrow$ uncertainty $\rightarrow$ explanation $\rightarrow$ retrieval $\rightarrow$ recommendation $\rightarrow$ safety $\rightarrow$ simulation $\rightarrow$ decision $\rightarrow$ human review, with deterministic routing to DEFER_DATA, DEFER_UNCERTAINTY, DEFER_EVIDENCE, REJECT_SCHEMA, REJECT_CONSTRAINT, REJECT_SIMULATION, PENDING_HUMAN_REVIEW or ERROR_TOOL. Only the recommendation node calls the LLM.

F. Safety constraints and simulation gate

Allowed actions are a bounded green extension (0–10 s) of an existing green phase, selection of a pre-validated program, or monitoring only. Thirteen deterministic checks (mapping validity, signal and phase existence, action type, bounds, minimum/maximum green, clearance preservation, pedestrian and side-road protection availability, data freshness, forecast reliability, evidence support) plus a text policy filter must pass. Candidates are then applied through a protective TraCI wrapper that refuses non-green phases and out-of-bound durations, and compared with the fixed-time baseline on matched seeds; a candidate passes only if corridor delay improves by at least 1%, side-road delay rises by at most 10% and no vehicle teleports. Each flow drives one fixed free-flow route so that demand stays on the corridor, the fixed-time programs are re-timed for the routine demand with Webster's method (SUMO `tlsCycleAdaptation.py`; a declared assumption, not NYC timing plans), and the testbed itself must first pass a baseline test: the side-street demand level is the largest at which the fixed-time baseline, on development seeds, has no teleport and at most 1% unfinished trips; without such a level no candidate is scored (NOT_EVALUATED). Pedestrian performance is not modelled and is reported as NOT_EVALUATED.

VI. Experimental Design

Partitions. The 5-minute time axis is split chronologically into training (60%), validation (15%), calibration (10%) and test (15%). A forecast origin belongs to a partition only if all of its targets fall inside it (purging); earlier history may be used as input. Scalers and link eligibility (training-partition coverage ≥ 0.5, lowered from the planned 0.6 after the coverage audit and recorded as such) use the training partition only.

Feature tracks. The *operational* track uses traffic, calendar, archived weather *forecasts* (Open-Meteo historical-forecast API; issue times are not exposed, a disclosed limitation) and 311 requests with a declared 24-hour publication lag. The *retrospective* track replaces forecasts by reanalysis weather and adds collisions and borough-level permitted events, whose operational availability cannot be established; it is used for contextual analysis only.

Metrics. MAE, RMSE, sMAPE and R^2 per horizon on a shared valid-target mask; paired day-block bootstrap for model differences; empirical interval coverage and width by horizon, congestion category and scenario; Recall@k, MRR and support-status accuracy for retrieval; schema validity, invalid-citation counts, constraint violations and decision distribution for advisories; SUMO time loss, travel time, waiting time, stops, queue length, throughput, side-road delay, unfinished trips, insertion backlog and teleports; and latency of each stage.

Training budget. Every deep model (GRU, temporal Transformer, proposed model, ablations) is trained with AdamW and early stopping as in Section V, under a recorded compute budget of 4,096 training windows per epoch, sampled afresh each epoch (`configs/training.yaml`); the budget, the resolved device and the stop reason are written into every run's configuration, and the budget is the same for all models.

Advisory cases and policies. Seven REPLAY cases chosen by rule from the test partition: routine rush hour (largest predicted drop at a weekday peak), rainy period (a wet weather code, present in this test partition), a stale-data stress test (inputs aged six steps), a wide-interval stress test (forced 99 mph width), an out-of-scope query (insufficient evidence), and two cases on the one link with a reviewed, high-confidence mapping to the simulated corridor (its largest predicted drop and an ordinary rush-hour origin). Policies: full system; LLM without retrieval; RAG without the constraint layer; RAG with constraints but without the simulation gate. The three stress

cases run under the full system only, giving 19 policy-by-case runs.

VII. Results

Every table in this section is generated by `python -m smartcity.cli paper sync` from the result files in `reports/experiments/`; a table reading NOT RUN means the corresponding stage has not produced an artifact and no number is claimed for it. The narrative is written only from those artifacts (`reports/verification/paper_claims.csv` maps every claim to its file).

A. What was actually run

All numbers in this section come from one real-data run of the released pipeline on the authors' workstation (Windows 11, RTX 4050 Laptop GPU with 6 GB, SUMO 1.28.0, Ollama with `qwen2.5:7b`), stage by stage, with the stage log, run ids and hashes recorded in `reports/verification/`. Of the 26 Manhattan DOT links, seven West Side Highway links report a speed of zero in every 2026 record (the feed's no-measurement encoding) and never enter the 5-minute grid; of the 19 gridded links, 16 pass the training-coverage rule and form the modelling set, and three of those (FDR N to the Triborough Bridge, Triborough Bridge westbound, BQE N to the Manhattan Bridge) have no valid observation inside the test partition and are reported as NOT_EVALUATED in `reports/experiments/link_test_support_operational.csv`. Of 1,968,303 raw speed records, 978,018 carry a zero speed and 59 exceed the plausibility bound; 990,226 are valid and the gridded observed fraction is 0.611. The chronological partitions over the 84,084 grid steps are 60/15/10/15 %: calibration covers 2026-08-08 04:20 to 09-06 08:55 UTC and the test partition 2026-09-06 09:00 to 2026-10-20 04:00 UTC (45 days; 144,217, 144,213 and 144,205 valid targets at 15, 30 and 60 minutes, identical for every model).

To finish inside a fixed compute deadline, every deep model was trained with a recorded budget of 4,096 training windows per epoch, sampled afresh each epoch without replacement from the 50,416 training origins (about 8 % of them), with early stopping on validation MAE (patience 5, at most 30 epochs) and one seed. The budget is identical for the GRU, the temporal Transformer, the proposed model and its ablations, and is stated here because it bounds how far each model was trained, not because it is a tuned hyper-parameter. Under it the proposed model stopped after 14 epochs (best epoch 8, 2.3 minutes on the GPU, 158,403 parameters), the temporal Transformer after 14 (best 8, 1.8 minutes), the GRU after 17 (best 11, 6.0 minutes, 41,667 parameters) and the no-weather ablation ran the full 30 epochs (best 24).

B. Multi-horizon forecasting

Table II gives the test error on the operational track. Every learned model beats persistence and the historical average at every horizon, and the three sequence models beat the two gradient-boosted baselines on MAE at every horizon. The proposed context-gated GATv2–Transformer has the lowest MAE at 30 and 60 minutes (5.33 and 6.15 mph against 5.34/6.18 for the temporal Transformer and 5.37/6.17 for the GRU), the GRU the lowest at 15 minutes (4.72 mph against 4.77 for the proposed model), and XGBoost the lowest 60-minute RMSE (8.98 against 9.05). Relative to persistence the proposed model reduces MAE by 7 %, 12 % and 16 % at the three horizons and raises R^2 from 0.55/0.46/0.30 to 0.69/0.63/0.56. The margins between the three deep models are 0.01 to 0.05 mph, which a single seed cannot separate, so this paper does not claim that the proposed architecture forecasts better than a well-trained GRU or temporal Transformer on this sparse 16-link graph; it claims that it is at least as accurate while carrying the availability gate and the graph layer that the rest of the workflow uses. On the retrospective track (reanalysis weather plus collisions and permitted events, contextual analysis only) the proposed model reaches 4.67, 5.36 and 6.13 mph (Table II is the operational table; the retrospective rows are in `forecast_metrics.csv`), a small gain at 15 and 60 minutes over the operational 4.77/5.33/6.15, with XGBoost and LightGBM at 4.89–4.90, 5.61–5.65 and 6.30–6.39.

Ablations. Table III reports the mean MAE over the three horizons and a paired day-block bootstrap (1,000 resamples of the 44 complete test days, 95 % percentile interval) of the difference to the proposed model (5.419 mph). Removing the availability gate (plain concatenation) raises MAE by 0.067 mph ($[0.048, 0.086]$); removing the 311 infrastructure group raises it by 0.014 mph ($[0.002, 0.026]$); removing the operational weather forecasts *lowers* it by 0.046 mph ($[-0.069, -0.021]$). The last is a negative result: with this data, this budget and the re-forecast weather proxy, the weather features hurt. The no-events and no-incidents variants are NOT_APPLICABLE on the operational track, where those groups do not exist, and are listed as such rather than as ablations. An earlier execution of this stage had resampled 5-minute steps instead of days (a unit error in the day index) and produced far narrower intervals; it was found during the run, corrected in code, re-run, and the superseded file is kept with the run's local evidence.

C. Uncertainty calibration and the reliability gate

Split-conformal half-widths fitted on 102,301 calibration residuals per horizon are 12.1, 13.5 and 14.8 mph. On the test partition the 90 % intervals reach empirical coverage of 0.909, 0.907 and 0.903 (Table IV), so the marginal guarantee holds, but the intervals are wide (mean widths 24.3, 26.9 and 29.5 mph on links whose training means lie between 16 and 32 mph). Conditional coverage is uneven: 0.93 on free-flowing targets (speed ratio ≥ 0.8, $n \approx 98{,}500$), 0.93 on moderate ones, and 0.80, 0.79 and 0.76 at the three horizons under congestion (ratio < 0.5, $n \approx 23{,}300$); 0.93–0.94 in rush hours against 0.90 outside them; 0.92–0.93 under wet weather codes ($n \approx 7{,}900$) against 0.90–0.91 when dry. The reliability gate, whose thresholds are fixed on calibration and validation data (interval width at most

TABLE I
VERIFIED PUBLIC DATA SOURCES (SERVER-SIDE AUDIT)

Source	Dataset	Observed range	Records	Role/limitation
speeds	i4gi-tjb9	1930-12-09 – 2026-10-20	112832791	Historical retention verified by server-side counts; includes outlier
volume	7ym2-wayt	–	–	Sparse sampled ATR counts; not continuous observed volume; used for SU
events	tvpp-9vvx	2024-06-30 – 2027-10-27	24837	Current feed only; does not reconstruct what was announced in the past
events_hist	bkfu-528j	2007-12-31 – 2026-12-27	2906868	Free-text locations (no coordinates); permit announcement time unknown
collisions	h9gi-nx95	2012-07-01 – 2026-06-11	2269187	Publication lag of months (latest record far behind today); not a live
sr311	erm2-nwe9	2020-01-01 – 2026-10-19	22854982	Creation time is not publication time; used with a declared 24 h lag.
weather		–	–	Reanalysis is not street-level measurement; archived forecasts lack is
osm		–	–	Current topology applied to the 2026 study period.

TABLE II
TEST-PARTITION FORECAST ERROR (MPH) BY MODEL AND HORIZON, OPERATIONAL FEATURE TRACK. PERIOD 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 UTC; IDENTICAL VALID-TARGET MASK FOR ALL MODELS (N PER HORIZON SHOWN).

Model	15 min		30 min		60 min	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
persistence	5.14	9.14	6.06	10.05	7.29	11.43
historical_average	7.63	10.48	7.63	10.49	7.64	10.49
xgboost	4.82	7.62	5.58	8.35	6.31	8.98
lightgbm	4.87	7.72	5.60	8.35	6.32	9.03
gru	4.71	7.55	5.37	8.26	6.17	9.03
temporal_transformer	4.73	7.62	5.34	8.31	6.18	9.12
proposed	4.77	7.63	5.33	8.28	6.15	9.05
n	144217		144213		144205	

TABLE III
ABLATIONS OF THE PROPOSED MODEL (TEST MAE, MPH, MEAN OVER HORIZONS)

Variant	MAE	Δ	Status
proposed	5.42	0.00	REFERENCE
ablation_no_gate	5.49	0.07	OK
ablation_no_weather	5.37	-0.05	OK
ablation_no_infrastructure	5.43	0.01	OK
ablation_no_events	–	–	NOT_APPLICABLE (no events group in operational track)
ablation_no_incidents	–	–	NOT_APPLICABLE (no incidents group in operational track)

TABLE IV
SPLIT-CONFORMAL INTERVALS: EMPIRICAL TEST COVERAGE (NOMINAL 90%) AND WIDTH

Horizon	n	Coverage	Width (mph)
15 min	144217	0.909	24.26
30 min	144213	0.907	26.95
60 min	144205	0.903	29.54

38.1 mph, the 80th percentile of validation widths; at most 34 % missing recent inputs; data at most 20 minutes old; at least 200 calibration samples), defers 25.5, 25.7 and 26.1 % of test forecasts at the three horizons; the retained forecasts have MAE 3.72, 4.38 and 5.25 mph against 4.77, 5.33 and 6.15 for all forecasts, and coverage 0.95, 0.94 and 0.93 (`artifacts/calibration/deferral_tradeoff.csv` tabulates the full trade-off from 0 to 100 % tolerated missing inputs).

TABLE V
RETRIEVAL EVALUATION ON THE LABELLED QUERY SET (LABELS DRAFTED BY THE PROJECT; REVIEW STATUS: DRAFTED_BY_PROJECT_UNREVIEWED)

Queries	28
Recall@8	0.818
MRR	0.702
Support-status accuracy	0.964

D. Explanations of the trained model

Exact grouped KernelSHAP (all 2^P coalitions, $P = 5$ or 6 players, 528 or 1,040 model calls and 1.7 to 4.2 s per explanation on the CPU, additivity gap below 10^{-3} mph) was computed on the trained proposed model for 12 test points: the six largest predicted 30-minute speed drops, one per link, and six random points, all drawn from the 132,355 of 201,616 test (origin, link) pairs whose link had a raw observation at every one of the last six input steps and an observed target (the eligibility rule introduced after a first pass had selected a link whose feed had been dark for four weeks). Averaged over the 12 points, the mean absolute attribution is 8.22 mph for the target link's own last 30 minutes, 2.04 for its earlier 90 minutes, 1.09 for calendar, 0.52 for graph-neighbour traffic, 0.19 for weather and 0.09 for 311 requests. In the largest alert, the George Washington Bridge lower-level plaza link at 30 minutes, the model predicts 2.25 mph against a 26.9 mph background and an observed 1.24 mph, with -21.3 mph attributed to the link's own recent traffic; on FDR N 25th–63rd St it predicts 1.92 mph (observed 6.83) with

TABLE VI
ADVISORY WORKFLOW OUTCOMES BY POLICY (REPLAY CASES; SIMULATION-ONLY EFFECTS)

Policy	Case	Decision	LLM	Invalid cit.	Constraint viol.
full_system	routine_rush_hour	REJECT_CONSTRAINT	OK	0	1
full_system	rain_period	DEFER_UNCERTAINTY	–	0	0
full_system	stale_data_stress	DEFER_DATA	–	0	0
full_system	wide_interval_stress	DEFER_UNCERTAINTY	–	0	0
full_system	insufficient_evidence	DEFER_EVIDENCE	–	0	0
full_system	mapped_corridor_alert	PENDING_HUMAN_REVIEW	OK	0	0
full_system	mapped_corridor_rush_hour	PENDING_HUMAN_REVIEW	OK	0	0
llm_without_retrieval	routine_rush_hour	REJECT_CONSTRAINT	OK	1	1
llm_without_retrieval	rain_period	DEFER_UNCERTAINTY	–	0	0
llm_without_retrieval	mapped_corridor_alert	REJECT_CONSTRAINT	OK	1	1
llm_without_retrieval	mapped_corridor_rush_hour	REJECT_CONSTRAINT	OK	1	1
rag_without_constraints	routine_rush_hour	PENDING_HUMAN_REVIEW	OK	0	1
rag_without_constraints	rain_period	DEFER_UNCERTAINTY	–	0	0
rag_without_constraints	mapped_corridor_alert	PENDING_HUMAN_REVIEW	OK	0	0
rag_without_constraints	mapped_corridor_rush_hour	PENDING_HUMAN_REVIEW	OK	0	0
rag_constraints_without_simulation	routine_rush_hour	REJECT_CONSTRAINT	OK	0	1
rag_constraints_without_simulation	rain_period	DEFER_UNCERTAINTY	–	0	0
rag_constraints_without_simulation	mapped_corridor_alert	PENDING_HUMAN_REVIEW	OK	0	0
rag_constraints_without_simulation	mapped_corridor_rush_hour	PENDING_HUMAN_REVIEW	OK	0	0

TABLE VII
SUPERSEDED 2026-10-20 CONTROLLER COMPARISON (SIMULATION), KEPT AS RECORDED: BARE FROM/TO FLOWS THAT SUMO ROUTES AT INSERTION ON CURRENT TRAVEL TIMES, THE NORTHBOUND FLOW INSERTED ON A 0.2 M STUB EDGE, DECLARED DEMAND. THE TESTBED AUDIT (TABLE VIII) SHOWS THAT THIS DESIGN SENT ABOUT HALF OF THE MAINLINE OFF WEST STREET, SO THE TABLE IS NOT A CORRIDOR CONTROLLER COMPARISON. MEAN OVER MATCHED SEEDS. SCENARIO-BASED TESTBED WITH OSM GEOMETRY AND DECLARED DEMAND, NOT A CALIBRATED DIGITAL TWIN. TIME LOSS IS THE MEAN PER *completed* TRIP; COMPLETED, UNFINISHED AND TELEPORTED VEHICLES ARE PER RUN, AVERAGED OVER SEEDS, SO A LOWER TIME LOSS WITH FEWER COMPLETED TRIPS IS SURVIVORSHIP, NOT IMPROVEMENT.

Scenario	Controller	Seeds	Loaded	Completed	Unfinished	Teleports	Time loss (s)	Corridor TL (s)	Side TL (s)
event_surge	actuated	5	8350	2877	4957	1667	1118.4	1877.9	965.1
event_surge	fixed_time	5	8350	2087	5856	1799	780.5	836.6	774.6
event_surge	forecast_rule	5	8350	2014	5955	1834	739.0	715.3	741.9
incident_reduced_capacity	actuated	5	8350	2240	5604	1971	1070.1	1816.3	946.1
incident_reduced_capacity	fixed_time	5	8350	1384	6566	2268	898.4	1405.6	863.5
incident_reduced_capacity	forecast_rule	5	8350	1236	6725	2174	806.0	1117.3	786.2
routine_peak	actuated	5	6675	2563	3682	1399	987.5	1470.8	847.3
routine_peak	fixed_time	5	6675	2358	3924	1330	971.6	1647.9	829.6
routine_peak	forecast_rule	5	6675	2358	3924	1330	971.6	1647.9	829.6

-13.9 from recent traffic and -3.2 from the earlier window. One selected point is a clear miss: FDR S Catherine Slip to the Brooklyn Bridge predicted 27.1 mph where 2.48 was observed, a drop not foreseeable from the inputs. Shortening the input history confirms the attribution: on a 20,665-target test subset the proposed model's MAE rises from 5.43 mph with the full two hours to 6.00 (60 min), 6.29 (30), 6.54 (15) and 7.11 (5 min). Removing all edges touching a link's graph neighbours changes its predictions by 1.34 mph on average for the four-neighbour Brooklyn Bridge–FDR N link (MAE 8.57 to 8.58), by 0.91 and 0.87 for FDR S 25th St–Catherine Slip and FDR N Catherine Slip–25th St (MAE 5.27 to 5.17 and 8.13 to 7.96, i.e. slightly better without the edges) and by 0.67 for the Brooklyn–Battery Tunnel eastbound approach, and by exactly 0.0 for West St S at Spring St, whose only remaining edges are outgoing (its upstream neighbour is a dead feed), so no message reaches it. The LightGBM TreeSHAP baseline attributes 65 % of its mean absolute attribution at 15 minutes to traffic lags, 16 % to calendar and 4 % each to weather and link identity. All of these are model sensitivities, not causal effects.

E. Evidence retrieval

The approved corpus holds 15 documents (the FHWA 2008 Signal Timing Manual and its chapters 5 and 6, MUTCD 2009 chapters 4D and 4E, nine SUMO documentation pages pinned to v1.28.0 and the project-defined policy), 1,633 chunks embedded with all-MiniLM-L6-v2 into a FAISS inner-product index; 171 chunks exceed the encoder's 256-token window and are truncated by it (recorded in `index_manifest.json`). On the 28 project-labelled queries the index reaches Recall@8 of 0.818 and MRR of 0.702, and the support-status decision (SUPPORTED against INSUFFICIENT_EVIDENCE at a 0.35 score threshold) matches the label on 27 of 28 queries (0.964, Table V). The labels were drafted by the project and have not been reviewed externally.

TABLE VIII
SUMO TESTBED AUDIT (SIMULATION). TOP: ROUTINE PEAK AT THE 2026-10-20 DECLARED DEMAND (1200/150 VEH/H PER DIRECTION/SIDE APPROACH), NETCONVERT PROGRAMS, SEED 1. BELOW: PREREGISTERED SIDE-DEMAND SELECTION ON THE ROUTINE PEAK, FIXED-TIME ONLY, OBSERVED ATR MAINLINE DEMAND, THREE DEVELOPMENT SEEDS, MEAN PER RUN; A LEVEL IS ADMISSIBLE ONLY WITH NO TELEPORT AND AT MOST 1 % UNFINISHED IN EVERY SEED. SELECTED LEVEL: NETCONVERT PROGRAMS: NONE; WEBSTER RE-TIMED PROGRAMS: NONE.

Route choice	Inserted	Teleports	Re-routed	Main off West St
dynamic_insertion	5690	680	1853	1120/2189
fixed_free_flow	4225	1044	0	0/1452
netconvert programs				
Side (veh/h/approach)	Loaded	Unfinished	Teleports	Time loss (s)
150.0	7085	4308	1018	1299.2
112.5	6060	3630	859	1023.6
75.0	5060	2510	565	961.7
37.5	4035	942	49	832.2
0.0	3010	533	0	803.6
Webster re-timed programs				
Side (veh/h/approach)	Loaded	Unfinished	Teleports	Time loss (s)
150.0	7085	1335	47	552.9
112.5	6060	1000	49	541.4
75.0	5060	641	42	564.9
37.5	4035	353	18	589.9
0.0	3010	130	0	591.2

F. Advisory workflow

Table VI lists all 19 policy-by-case runs (REPLAY mode on test-partition forecast origins). The deterministic gates fired as designed under every policy: the stale-data stress case (inputs aged six steps) ended in DEFER_DATA, the wide-interval stress case (forced 99 mph width) and the observed rainy period, whose intervals of 44–46 mph exceed the 38.1 mph cap, in DEFER_UNCERTAINTY, and the out-of-scope query in DEFER_EVIDENCE, each without an LLM call. The model was called 12 times (one attempt each, 1,790 to 3,604 prompt tokens, 181 to 258 completion tokens, 16 to 126 s per call with the first, cold call slowest) and returned schema-valid JSON every time (`reports/experiments/llm_calls.csv`). Without retrieval it cited an evidence id that does not exist in all three cases (for example `EVIDENCE_123456`), and the citation check rejected all three; with retrieval, no invalid citation occurred in nine calls. On the routine rush-hour case (FDR S Catherine Slip–Brooklyn Bridge, a link without a reviewed DOT-to-SUMO mapping) the constraint layer rejected the candidate under every policy that keeps it, while the RAG-without-constraints policy passed the same candidate to human review. On the two cases on the mapped West Street link the model recommended monitoring only, which passes all 13 checks (pedestrian protection NOT_EVALUATED) and ends in PENDING_HUMAN_REVIEW; because no signal change was proposed, the simulation gate was never invoked by a real advisory and the SIMULATION_PASSED path remains exercised only by the software tests. End-to-end latency is 19 to 30 s per advisory after the first call.

G. Simulation testbed

The West Street corridor extracted from OpenStreetMap has 82 corridor edges, 22 signalised intersections and 25 side-street entries. Two DOT links map to the corridor with high confidence: a dead West Side feed with no usable speed data (11th Ave S Gansevoort–West St at Spring St) and the modelled West St S Spring St–Brooklyn–Battery Tunnel portal link. The scenarios' demand and congestion flags are declared rather than derived from that link's observed or forecast speeds, so the testbed is a transfer demonstration on a real corridor rather than a digital twin of a forecast link. Pedestrian performance is NOT_EVALUATED because the scenarios contain no pedestrian demand.

First comparison (2026-10-20, superseded). With 1,200 vehicles per hour and direction on the corridor and 150 per side approach, both declared, and three scenarios (routine peak; event surge, demand $\times 1.25$; an incident that at the same surge caps one corridor edge at 2.5 m/s for 20 minutes), all 45 matched-seed runs (fixed-time, actuated and forecast-informed-rule controllers, five seeds, 300 s warm-up and 3,600 s evaluation, SUMO 1.28.0, 9,645 s of wall time) completed (Table VII). The corridor was oversaturated under every controller: of 6,675 (routine) or 8,350 (surge and incident) loaded vehicles per run only 1,003 to 3,441 completed their trip, 3,257 to 6,979 remained unfinished and 1,091 to 2,386 were teleported after waiting 300 s. The forecast rule's lower time loss per completed trip in the two stress scenarios (739.0 and 806.0 s against 780.5 and 898.4 for fixed-time) came with fewer completed trips (2,014 against 2,087; 1,236 against 1,384), i.e. survivorship, and we claimed no controller ranking.

Testbed audit (2026-10-21). To find out why, the corridor was rebuilt in the development container from the OSM editing API (Overpass is blocked there) for the box 40.700–40.737 N, 74.000–74.020 W with the same highway filter. It reproduces the workstation corridor exactly (the same 82 corridor edges, 22 signals, corridor phases, entry and exit edges and both DOT mappings; 24 of 25 side entries keep their edge id, the 25th is the same FDR Drive approach under another OSM way id), and the original design run on it reproduces the oversaturated regime (routine peak, fixed-time, seed 1: 2,981 completed, 3,294 unfinished and 928 teleports, against 2,746, 3,527 and 1,157 on the workstation). The audit found two defects behind Table VII. First, demand was written as bare from/to flows, which SUMO routes at insertion on the current travel times [18]: at the declared demand the rerouting device replaced 1,853 routes and sent 1,120 of 2,189 mainline vehicles off West Street into the Lower Manhattan grid (Table VIII), so the 45 runs did not compare controllers on the corridor. Second, the northbound chain began on a 0.2 m stub left by junction joining, which throttled northbound insertion. Both are corrected (one fixed free-flow route per flow; the entry moves to the first corridor edge of at least 20 m), and the mainline demand now comes from the only weekday NYC automated traffic count on the corridor, northbound at

Gansevoort Street on 6–10 October 2014 (15 complete PM-peak hours, median 1,389 veh/h), applied to both directions; the only southbound count (Bloomfield Street, 7–8 March 2015, median 2,074 veh/h) fell on a weekend and is excluded by the weekday rule. Fixed routes make the corridor more congested, not less (1,044 teleports against 680 at the declared demand), because traffic can no longer leave it.

Corrected protocol: comparison not run. Side-street demand has no matched count, so its level was to be chosen by a rule fixed in `configs/simulation.yaml` before any comparison: on the routine peak, under the fixed-time baseline only and on three development seeds that the comparison does not use, the largest fraction of the declared 150 veh/h at which every seed has no teleport and at most 1 % of loaded vehicles unfinished. With the programs netconvert generates, no level qualifies (Table VIII): even with no side-street demand at all, 533 of 3,010 loaded vehicles per run remain unfinished and the mean time loss is 804 s. Counting only the movements each mainline route makes, the guessed programs give the northbound mainline 25 s of a 90 s cycle on two lanes at one signal, about 1,000 veh/h at a nominal saturation flow of 1,800 veh/h per lane and below the observed 1,389 veh/h (`simulation_signal_capacity.csv`). We therefore re-timed the fixed-time programs for the demand being tested with Webster's method as implemented in SUMO's `tlsCycleAdaptation.py` (tool defaults; phase order and states unchanged), a declared assumption fixed before any comparison under this protocol, and applied the same selection rule. Re-timing raises every mainline signal's estimate to at least 1,800 veh/h northbound and 2,040 southbound and removes most of the saturation: with the full declared side demand 5,220 to 5,229 trips complete per run instead of about 2,300, and with no side demand teleports stay at zero and the northbound mainline completes all 1,389 trips at 195 to 197 s of time loss. No level qualifies, however: even without side demand 129 to 131 vehicles per run (4.3 %) remain unfinished, and the southbound mainline loses 1,014 to 1,038 s per trip; green shares do not explain this and the audit does not identify its cause. Under the preregistered rule the corrected controller comparison is NOT RUN, and the advisory simulation gate returns NOT_EVALUATED (ERROR_TOOL with its cause) rather than judging a candidate on a testbed that cannot serve its own baseline. We did not relax the rule after seeing these results.

H. Runtime

Training times above are on the GPU; inference of one forecast origin for all links on the CPU costs 5.7 ms for the proposed model, 2.2 for the temporal Transformer, 0.7 for the GRU and about 0.1 for XGBoost; a FAISS query takes tens of milliseconds after the first (model-loading) call; the full per-link runtime table is `reports/experiments/runtime_metrics.csv`.

VIII. Discussion

The system is designed so that each component's contribution can be isolated: baselines and the proposed model share one test mask, ablations remove one context group or the gate at a time, and the advisory evaluation runs the same cases under four policies that switch off retrieval, the constraint layer or the simulation gate. Read together, the results support some of the proposals in Section I and not others.

Forecasting. The accuracy gain of the proposed model over persistence (7–16 % MAE) and over the tree baselines is real and consistent across horizons, but its margin over the GRU and the temporal-only Transformer is within what a single seed can resolve. Two measured facts explain why the spatial layer adds little here: the graph is sparse (16 nodes with usable data, most of them limited-access highway segments whose monitored neighbours are kilometres away) and the explanations attribute 0.52 mph of mean absolute effect to neighbour traffic against 8.22 mph to the link's own last half hour. The neighbour-removal analysis shows the layer is not inert (prediction changes of 0.7–1.3 mph on the connected links) but also that removing edges sometimes lowers the error, so on this network message passing is a modest and mixed contribution. The availability gate is the component the ablations do support: removing it costs 0.067 mph with an interval that excludes zero.

Context. The weather ablation is the clearest negative result: the operational re-forecast weather features raise MAE by 0.046 mph, and the retrospective track, which uses reanalysis weather, is only marginally better than the operational one. On a highway-heavy link set over nine months the weather signal reaching a 5-minute speed is weak relative to the traffic history, and a re-forecast proxy adds noise. The 311 infrastructure group helps by a small but non-zero amount. Whether the events and incidents groups help cannot be answered on the operational track, where the data to compute them were not available at prediction time.

Uncertainty. The conformal intervals hold their marginal 90 % guarantee but are wide and under-cover congested targets (0.76–0.80), which is exactly the regime an advisory cares about. The deterministic gate turns this into deferrals: a quarter of forecasts are withheld and the remainder are a third more accurate. For an operator this is the honest trade-off; a calibration conditional on congestion state is the obvious next step.

Grounding and constraints. The policy ablation isolates two mechanisms. Retrieval removes fabricated citations (three of three without it, none of nine with it), and the citation check catches them when they occur. The constraint layer, not the language model, decides admissibility: the same model output on an unmapped link is rejected with constraints and forwarded to human review without them. What the run does not show is a signal-timing recommendation surviving the whole chain into simulation: on the mapped corridor the model chose monitoring only, so the simulation gate's acceptance rule has been exercised by tests, not by a real advisory. We report

this as a gap rather than fill it with a hand-written candidate.

Simulation. The controller comparison is the clearest example of a result that a less careful reading would have turned into a gain: in the first 45 runs the forecast-informed rule shows 5 to 10 % lower time loss per completed trip in the two stress scenarios, but it completes fewer trips under an oversaturated corridor. Auditing the testbed instead of the controllers changed the diagnosis twice. The first suspect, the declared demand, was not the cause: the observed weekday West Street count (1,389 veh/h) is higher than the declared 1,200. What the audit found was a testbed that was not measuring the corridor at all, because SUMO's insertion routing sent half of the mainline through the Lower Manhattan grid, and, once routes were fixed, a corridor whose generated signal timing gives the northbound mainline about 1,000 veh/h where 1,389 are observed. Re-timing the programs with Webster's method serves the northbound flow but leaves a southbound delay the audit cannot explain, so the testbed still fails its own baseline. A controller comparison on such a testbed would rank signal plans by how they cope with modelling artefacts, so the preregistered selection rule stops it, and the advisory gate refuses to judge candidates on it. The lesson for simulation-gated advisory is that the gate's testbed needs its own acceptance test, run on the baseline alone, before any candidate is scored against it.

What is not claimed. Nothing here is a deployment result, a causal explanation, or a congestion reduction on a real street. The language model never holds safety authority (constraints and the simulation gate are code), retrieved text is untrusted evidence that must be cited by chunk id, and the most permissive status the workflow can reach is approval for simulation, never deployment.

IX. Limitations

Data. The DOT link set is sparse and highway-heavy; the full-year link enumeration timed out server-side, so the study uses the 26 Manhattan links active since 2026-09-20, of which seven are dead feeds, three more have under 3 % coverage, three lose their feed before or during the test partition (NOT_EVALUATED) and 16 are modelled; about half of all raw records are zero-speed no-measurement rows, treated as missing. Monitored segments do not tile the highways and the published polylines stop short of the named link ends, so the graph needs name-based continuity rules and remains sparse (16 directed edges, 6 isolated nodes before eligibility). The eligibility threshold was lowered from the planned 0.6 to 0.5 training coverage after the coverage audit and before any result on the 0.6 set existed; both are recorded. Operational weather is Open-Meteo's historical-forecast product, a re-forecast proxy rather than an archived agency forecast. Collisions lag publication by months and are retrospective only.

Modelling and evaluation. All deep models were trained under a recorded budget of 4,096 windows per epoch with early stopping, one seed and one chronological split; the only variance estimate is a day-block bootstrap on paired differences, and the proposed model's margin over the GRU and temporal Transformer is inside what seeds could change. Split conformal intervals carry a marginal guarantee under exchangeability, which traffic series violate; the measured coverage under congestion is 0.76–0.80. Explanations describe the trained model, with a background drawn from the training partition; neighbour sensitivity perturbs the graph in ways the model never saw during training, and only incoming edges can change a link's own prediction.

Knowledge, language model and agents. The approved corpus is small and is not NYC DOT's signal-timing policy; the project-defined policy is labelled as such. 171 of 1,633 chunks exceed the encoder's 256-token window. The retrieval labels were drafted by the project without external review, so recall and MRR are an internal consistency check. The language model is a 7B quantised model served locally; its advisories are only as good as the evidence and the constraints that follow it. The advisory evaluation is functional (seven cases, four policies, 19 runs), not statistical, and on the mapped corridor the model proposed no signal change, so the simulation acceptance path was not exercised by a real advisory. Two defects were found during the run and fixed before the reported numbers were produced (the bootstrap day index, and a missing field that had silently prevented every LLM call in a first agent pass); both superseded runs are kept as local evidence and described in `progress.md`.

Simulation. The SUMO testbed is a scenario built from OSM geometry with declared demand and default signal programs, not a calibrated digital twin, and contains no pedestrian demand; only one of the two mapped DOT links is a forecastable one and the scenarios' demand and congestion flags are declared rather than derived from its observations, so the testbed is a transfer demonstration. The first 45-run comparison used SUMO's insertion routing and a stub entry edge and is superseded. Its corrected protocol uses the only weekday automated count on the corridor, which is from 2014, twelve years before the speed data, and covers one direction. The signal programs are generated by netconvert or re-timed by Webster's method, not observed NYC timing plans; with either, the corrected testbed fails its baseline test even without side-street demand, so no admissible demand level exists and the corrected controller comparison is not run; the cause of the remaining southbound delay is not identified. No simulated gain is claimed. *Engineering.* Data download, training, LLM inference and the first simulation runs were executed on the authors' workstation (torch 2.11.0+cu128, a recorded deviation from the lock file). On 2026-10-20 the development container could not reach the data portals or the model server; on 2026-10-21 it could reach NYC Open Data, Crossref and the OSM editing API (not Overpass), and the testbed audit ran there with the same SUMO version. The container images were built and smoke-tested on the workstation only. Model weights, prediction arrays, raw data and the FAISS index are not in the repository; their hashes and run ids are. The complete, maintained list is in `reports/limitations.md`.

X. Conclusion and Future Work

We presented a complete, reproducible prototype that joins multi-horizon graph-Transformer forecasting, conformal calibration, model-faithful explanations, allow-listed retrieval, a locally hosted language model constrained by deterministic checks, and matched-seed simulation into one auditable advisory workflow on public New York City data, and we ran it end to end on nine months of real DOT speeds. The measured outcome is mixed and reported as such: the proposed forecaster beats persistence by 7–16 % MAE and the tree baselines at every horizon, but not the GRU or temporal Transformer by a margin a single seed can establish; the availability gate helps and the operational weather features hurt; conformal intervals hold their marginal coverage but under-cover congestion; retrieval removes fabricated citations and the constraint layer, not the model, decides admissibility; no advisory reached the simulation gate because the model chose monitoring on the mapped corridor; and no controller comparison is established, because an audit of the simulation testbed found that the first comparison let SUMO divert half of the corridor traffic and that, once corrected, the modelled corridor cannot serve the observed West Street volume within its baseline test, with generated or Webster re-timed signal programs. Future work, in the order the evidence suggests, is multi-seed training without the compute budget, congestion-conditional calibration, a denser sensor network on which the spatial layer can matter, reviewed signal timing plans and recent continuous counts for the simulation testbed, pedestrian modelling, and an externally reviewed retrieval benchmark.

References

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *International Conference on Learning Representations (ICLR)*, 2018, arXiv:1707.01926.

[2] B. Yu, H. Yin, and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *Proc. IJCAI*, 2018, pp. 3634–3640.

[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph wavenet for deep spatial-temporal graph modeling," in *Proc. IJCAI*, 2019, pp. 1907–1913.

[4] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *International Conference on Learning Representations (ICLR)*, 2022, arXiv:2105.14491.

[5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998–6008, arXiv:1706.03762.

[6] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[7] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021, arXiv:2107.07511 (preprint).

[8] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *The Annals of Statistics*, vol. 51, no. 2, pp. 816–845, 2023.

[9] C. Xu and Y. Xie, "Conformal prediction interval for dynamic time-series," in *Proc. 38th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 11 559–11 569.

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, arXiv:1705.07874.

[11] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, arXiv:2005.11401.

[12] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[13] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[14] Qwen Team, "Qwen2.5 technical report," 2024, arXiv:2412.15115 (preprint).

[15] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using SUMO," in *Proc. 21st IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2018, pp. 2575–2582.

[16] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," 2019, iCLR Workshop on Representation Learning on Graphs and Manifolds; arXiv:1903.02428.

[17] B. Rozemberczki, P. Scherer, Y. He, G. Panagopoulos, A. Riedel, M. Astefanoaei, O. Kiss, F. Beres, G. López, N. Collignon, and R. Sarkar, "PyTorch Geometric Temporal: Spatiotemporal signal processing with neural machine learning models," in *Proc. 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021, pp. 4564–4573.

[18] Eclipse SUMO contributors, "Automatic routing / routing device (SUMO 1.28.0 documentation)," https://github.com/eclipse-sumo/sumo/blob/v1_28_0/docs/web/docs/Demand/Automatic_Routing.md, 2026, section "Incomplete trips and flows; accessed 2026-10-21.